

Extreme Subgroup Effects

Under a Uniform Treatment Benefit

A Simulation Study Based on Observed Trial Data

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forestsearch R package · github.com/larry-leon/forestsearch

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*Before concluding a subgroup finding is real, compare it against what chance alone produces for a subgroup of **the same size**.*

The Central Question

Does an apparent subgroup finding reflect

genuine heterogeneity** or is it **noise amplified by small sample size?

What reviewers see

- Forest plots with dozens of subgroup HRs and 95 % CIs.
- Small subgroups routinely show dramatic point estimates and wide intervals.
- Country-level analyses, biomarker strata, and post-hoc cuts amplify the problem.

Why it matters

- Regulatory and HTA reviews increasingly scrutinize subgroups.
- Apparent “no-benefit” or “harm” signals in small groups can alter approval decisions.
- Clinical rationale does not automatically translate into statistical protection.

What chance alone does

- Even when the treatment helps every patient equally, small subgroups produce extreme estimates.
- Wide CIs in small N are the Cox model working correctly.
- Without calibration, “extreme” cannot be interpreted.

Strategy: Let the Trial Data Define the Null

Observed trial data

GBSG · N = 686 · all baseline covariates, censoring, outcomes

Weibull AFT model + super-population

generate_aft_dgm_flex() · fits the source structure

Synthetic trials with a uniform benefit

simulate_from_dgm() · true HR = 0.70 for every patient

The data-generating mechanism (DGM) is built from the trial itself

Observed covariate distributions, censoring process, and individual patient characteristics all carry over into the simulation.

Treatment effect is overridden to be **perfectly uniform**

$$\psi^0(L) = \log(HR_0) \text{ for all } L.$$

Any apparent subgroup finding is, by construction, **a false positive**.

Key design feature

The DGM is study-specific, not generic. Covariate correlations, event rates, censoring patterns — everything that makes real subgroups correlated with each other — is preserved. This is what separates the framework from abstract power calculations.

Case Study: the GBSG Breast Cancer Trial

686

patients with complete prognostic data

299

recurrence-free survival events

~33

months median event time

1984-89

enrollment period · Schumacher et al., JCO 1994

Key variables

hormon	treatment: hormonal therapy (0 / 1)
rfstime	recurrence-free survival time (days)
status	event indicator (0 = censored, 1 = event)
age	age in years
meno	menopausal status (0 = pre, 1 = post)
size	tumour size (mm)
grade	tumour grade (1, 2, 3)
nodes	number of positive nodes
pgr	progesterone receptor (fmol)
er	oestrogen receptor (fmol)

Setting: a Perfectly Uniform Treatment Effect

True log hazard ratio is constant across every patient:

$$\psi^0(L) = \log(\text{HR}_{\text{true}}) \quad \text{for every covariate profile } L.$$

HR = 0.70

A moderate treatment benefit — identical for every subject. Specified via the spline anchors:

`log.hrs = log(c(0.70, 0.70, 0.70))`

No embedded subgroup

`model = "null"` in `generate_aft_dgm_flex()`.

There is no harm subgroup, no effect modifier, no heterogeneity. Every subgroup shares exactly the same HR.

Any finding is a false positive

By construction. What we observe across simulated trials is the pure sampling-variability distribution of subgroup estimates.

Installing forestsearch & Accessing the Vignette

Installation

1. Install pak (one time)

```
install.packages("pak")
```

2. Install forestsearch from GitHub

```
pak::pak("larry-leon/forestsearch")
```

3. Load the package

```
library(forestsearch)
```

Documentation site

larry-leon.github.io/forestsearch

includes function reference and vignettes.

Extreme Subgroups vignette

larry-leon.github.io/forestsearch/articles/extreme_subgroups.html

Related vignettes on the pkgdown site

Getting Started

End-to-end walkthrough with the GBSG trial

Treatment Effect Definitions

Marginal HR, AHR, and CDE estimands

Methodology

Statistical framework, algorithm details, simulation results

Simulation Studies

Operating characteristics and benchmarks

Simulation Workflow

	Step	Function	Purpose
1	Prepare data	—	Convert source data to a consistent time scale (e.g. days → months)
2	Build the DGM	<code>generate_aft_dgm_flex()</code>	Fit a Weibull AFT model; create a super-population (N = 5 000)
3	Diagnose censoring	<code>check_censoring_dgm()</code>	Compare simulated vs. observed censoring rate and KM median
4	Calibrate censoring	<code>calibrate_cens_adjust()</code>	Find a log-scale shift so simulated censoring matches the trial
5	Simulate trials	<code>simulate_from_dgm()</code>	Draw synthetic RCTs from the calibrated DGM (one call per trial)
6	Fit Cox in subgroups	<code>coxph()</code> (stratified)	For each pre-defined subgroup, store HR point estimate and UB(HR)
7	Summarise	<code>summarize_extreme_sims()</code>	Empirical 99 % CI (ECI), $\Pr(\text{UB} \geq 1)$, conditional UB quantiles
8	Visualise	<code>plot_hr_forest()</code> / <code>plot_ub_forest()</code>	Forest plots with random-benchmark anchors by subgroup size

Subgroups Under Study — 56 in Total

ITT 1	Clinical 4	Continuous 13
All Patients (N ≈ 686)	Pre-/Post-meno · Grade 3 · Grade 1/2	Age · Size · Nodes · PGR · ER (various cuts)
2-way interactions 29	3-way interactions 5	Random benchmarks 4
Meno × Grade / Age / ER · Grade × Nodes / ER / PGR · Size × Nodes	Pre-meno/Young/ER-low · Post-meno/G3/Node+ · ...	random60 · random40 · random20 · random15 (re-sampled each sim)

The random* subgroups are the calibration anchor

They have no clinical meaning whatsoever — their true HR equals the ITT HR of 0.70. Any extreme finding is pure sampling variability. Placing them alongside clinical subgroups of similar size makes the noise floor explicit for each size class.

What Is Measured Per Subgroup, Per Simulation

Primary analysis

```
coxph(Surv(y_sim, event_sim) ~ treat_sim + strata(grade), data = subgroup_data)
```

HR point estimate distribution

Across all simulated trials, record the HR estimate for each subgroup.

Empirical 99 % CI (ECI)

median + 1st and 99th percentiles

$\Pr(\text{HR} < 0.80)$

frequency of “alarming” point estimate

Under truth the median should sit at 0.70 for every subgroup. The spread is what changes.

Upper 95 % CI bound UB(HR)

What a safety reviewer looks at when asking whether harm can be ruled out.

$\Pr(\text{UB} < 1.0)$

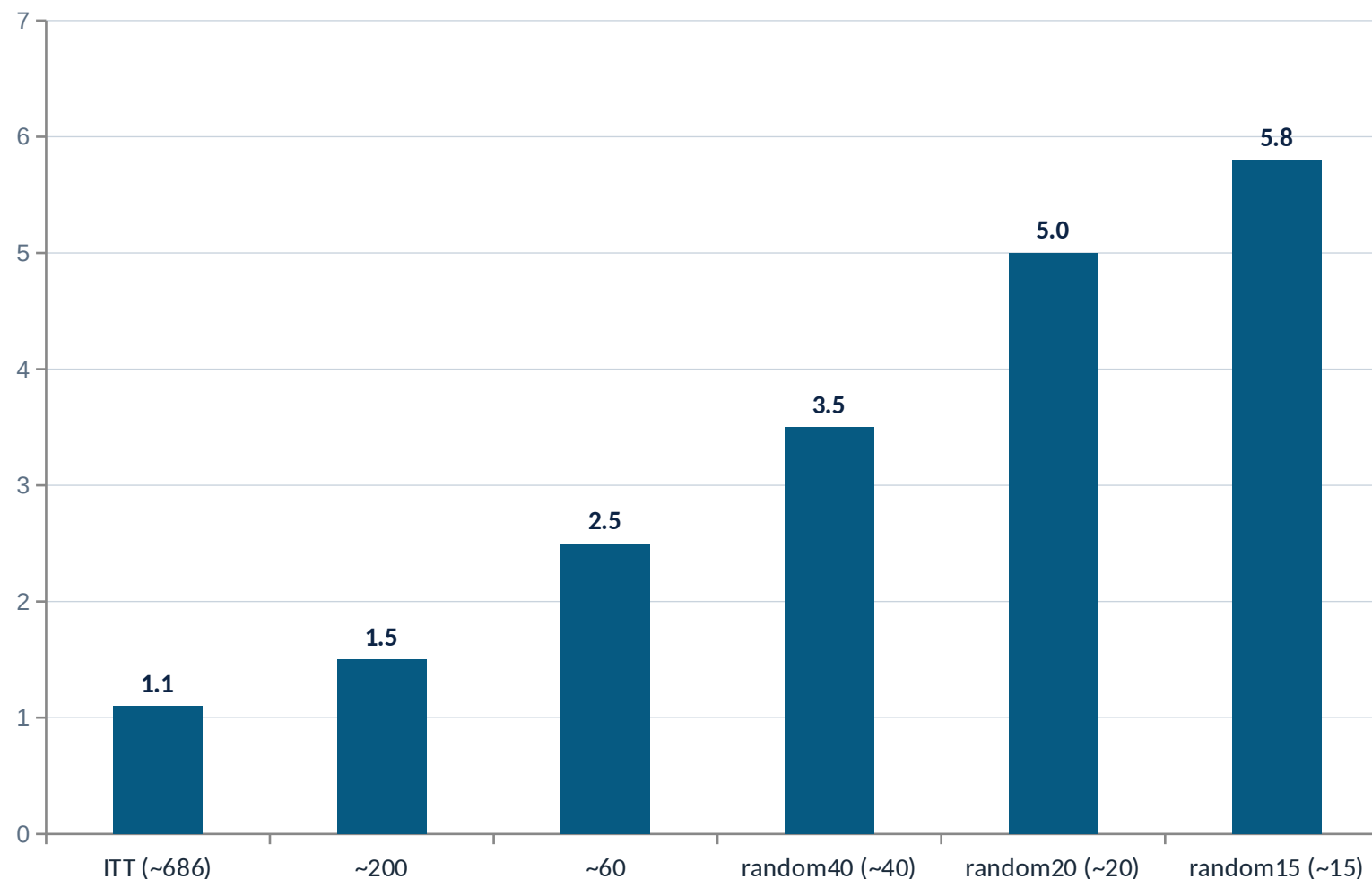
power to demonstrate benefit; drops sharply with N

Median UB | $\text{UB} \geq 1.0$

worst-case bound when the CI fails to exclude 1.0

The key safety quantity — how large the upper bound can grow by chance alone.

What Chance Alone Produces: 99th Percentile of UB(HR)



Illustrative quantitative anchors drawn from the null-DGM simulation summary.

Reading the chart

Each bar is the 99th percentile of the Cox 95 % upper confidence bound across thousands of simulated trials where the true HR is 0.70 everywhere.

In 1 % of trials

UB(HR) > 3.5 in a 40-patient subgroup.
UB(HR) > 5.0 in a 15-patient subgroup.

...even though the treatment benefits every patient equally.

Interpreting Subgroup Findings with the Benchmarks

Before concluding that a subgroup finding reflects real heterogeneity, compare it against what chance alone produces for a subgroup of the same size.

1

Wide CIs are correct

The Cox model is working exactly as intended. A 95 % CI from 0.3 to 4.5 in a 15-patient subgroup is a faithful summary of the evidence — not a bug.

2

Clinical rationale $\neq$ statistical protection

Clinical subgroups of $N \approx 40$ have the same HR / UB distribution as random benchmarks of $N \approx 40$. The naming and motivation do not reduce variability.

3

Calibrate, then interpret

A UB(HR) of 3.5 in a 40-patient subgroup is unremarkable against random40. The same value in a 300-patient subgroup is genuinely extreme. Context is everything.

Swap in Your Own Trial Data

The framework is designed so that GBSG is an example, not a requirement. Any trial with a time-to-event outcome, a treatment indicator, and baseline covariates can be plugged in.

Outcome	Treatment	Covariates	Subgroups
A time variable and an event indicator (0/1).	A binary treatment arm variable.	Any continuous or categorical baseline variables — age, biomarkers, stratification factors.	Your pre-defined subgroups of interest, expressed as logical / subset conditions.

What actually changes in the code

```
data          = my_trial_data      # replaces gbsg
outcome_var   = "time_months"      # your time column
event_var     = "event"            # your event indicator
treatment_var = "arm"              # your treatment indicator
continuous_vars = c("age", "biomarker", ...) # your continuous covariates
factor_vars   = c("region", "ECOG", ...) # your factor covariates
```

Six Concrete Steps to Adapt the Vignette

1 Align the time scale

Convert event times to the units the DGM will use (days → months is typical). Time-scale mismatch is the single most common cause of universal administrative censoring.

2 Swap the data call

Replace `data = gbsg` with your dataset. Map your column names into `outcome_var`, `event_var`, `treatment_var`, `continuous_vars`, `factor_vars`.

3 Set model = "null"

For the extreme-subgroup evaluation, use `subgroup_vars = NULL` and `model = "null"` so every patient shares the same true HR.

4 Redefine the subgroup list

Edit the subgroups list object. Use the same structure (`id / name / grp`) with expressions tailored to your covariates. Keep the `random15/20/40/60` benchmarks — they are the anchor.

5 Calibrate censoring

Run `check_censoring_dgm()` then `calibrate_cens_adjust()` so synthetic trials reproduce your study's censoring rate. Set `analysis_time` and `max_entry` to match your trial.

6 Run, summarise, plot

Use the same simulation loop: `simulate_from_dgm()` → `coxph()` → `summarize_extreme_sims()` → `plot_ub_forest()`. All tables and plots update automatically.

Tip • start by replacing only data and the column mappings; keep all tuning parameters at their defaults for the first run.

Conclusions

01**Small subgroups produce extreme results by chance**

Under a uniform HR = 0.70, subgroups with $N \lesssim 40$ regularly yield $UB(HR) > 3.0$ — purely from sampling variability.

02**Clinical rationale offers no statistical protection**

Clinical and random subgroups of the same size have statistically indistinguishable HR and UB distributions.

03**The Cox model is correct**

Wide intervals in small subgroups are faithful reflections of genuine uncertainty, not a modelling failure.

04**Simulation-based calibration is essential**

Without a null-DGM benchmark, “extreme” cannot be meaningfully quantified for a given subgroup size.

05**The framework is portable**

Any trial can be swapped in for GBSG. The forestsearch package provides the full, reproducible pipeline.

Thank you

forestsearch is an open-source R package providing a reproducible pipeline for simulation-based subgroup calibration.

Package

```
pak::pak("larry-leon/forestsearch")  
  
library(forestsearch)
```

Docs & vignette

larry-leon.github.io/forestsearch
[.../articles/extreme_subgroups.html](https://larry-leon.github.io/forestsearch/articles/extreme_subgroups.html)